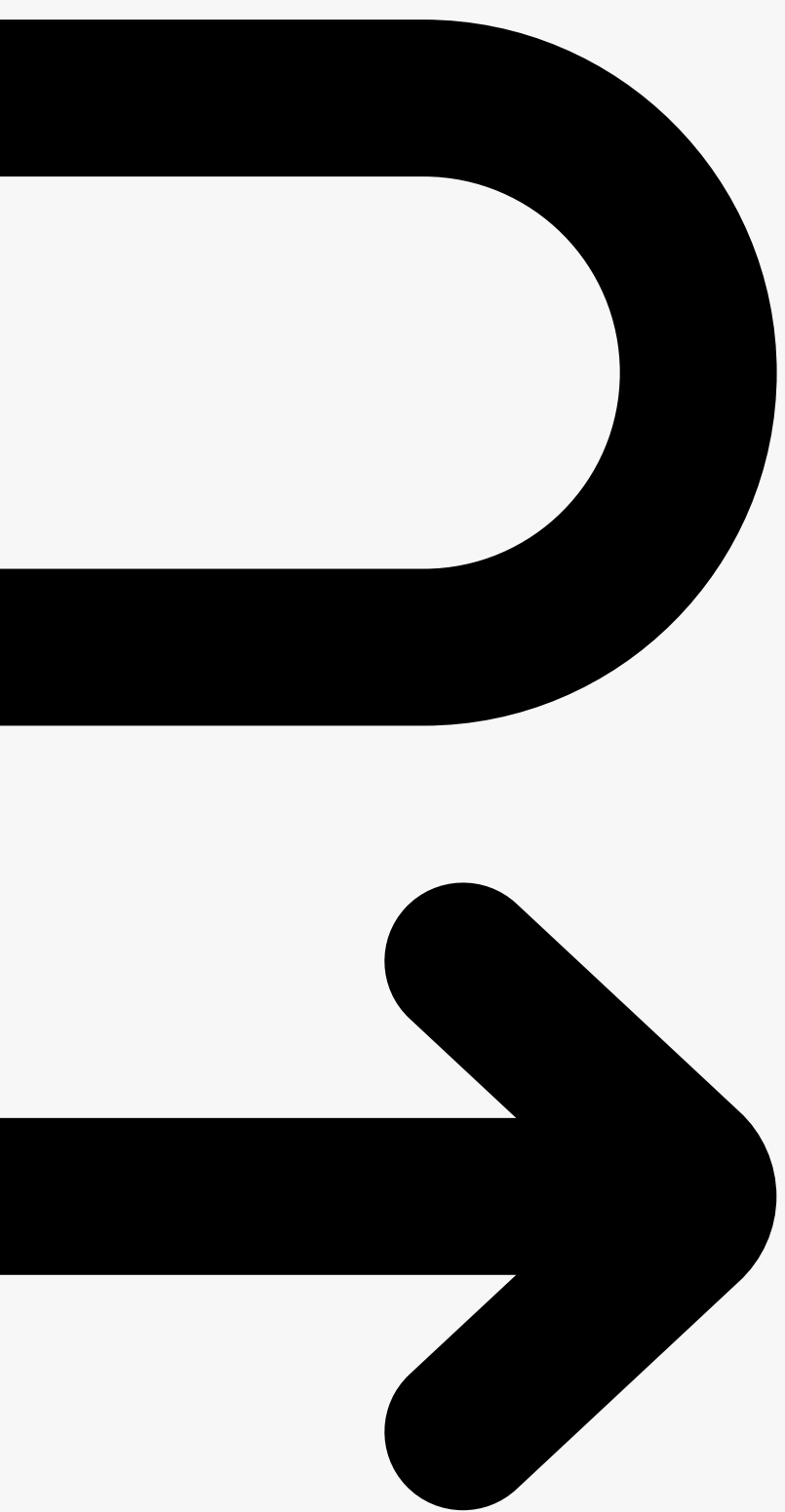
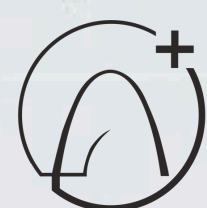




VISION TRACKING

FOR URBAN DESIGN NEEDS



Morphotect



دانشگاه تهران

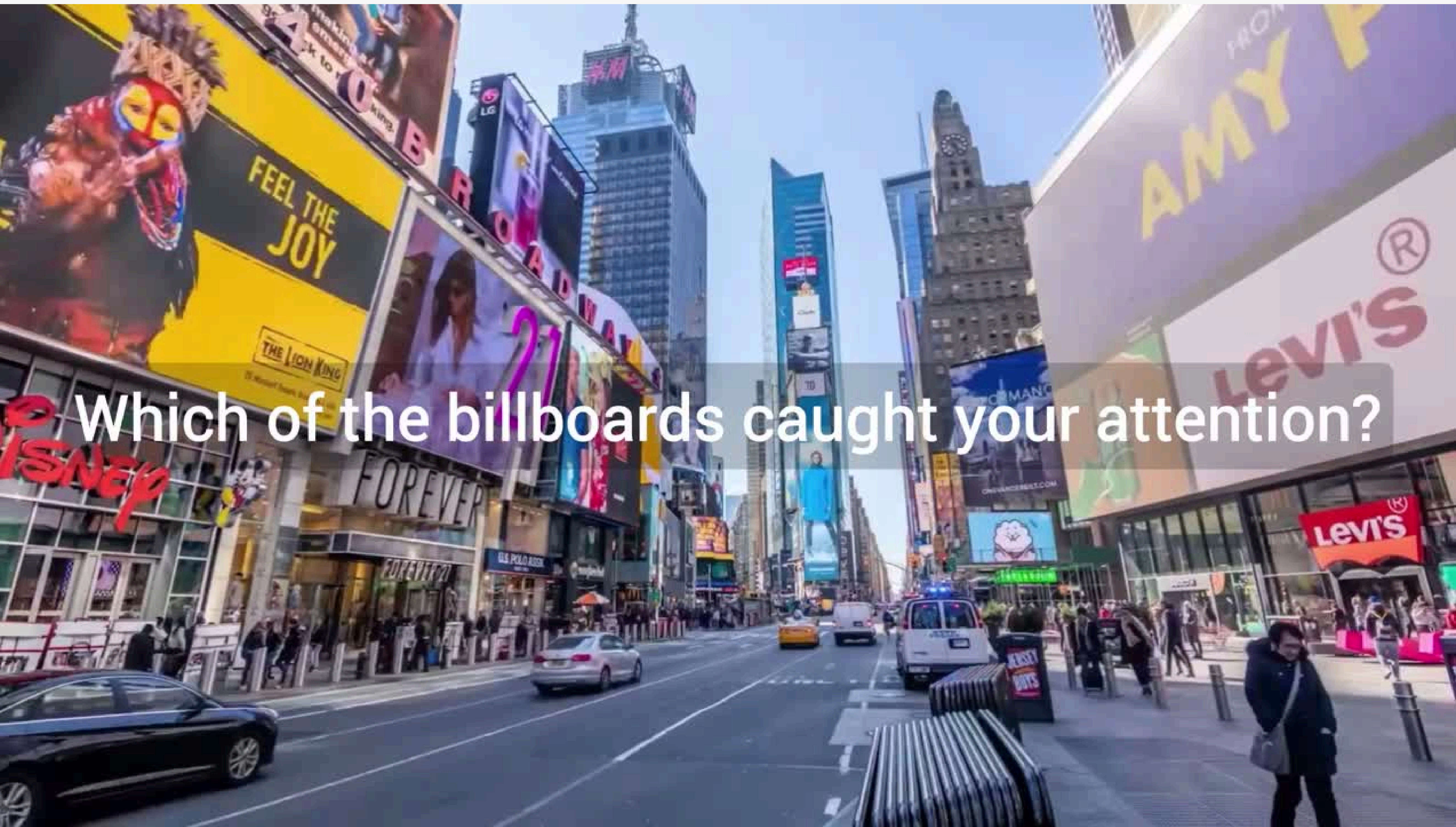
TEAM MEMBERS



We are a group of three:

- Parham Salahi,
- Mobina Mohseni,
- Mahdieh Moradi





Which of the billboards caught your attention?

PROBLEM STATEMENT

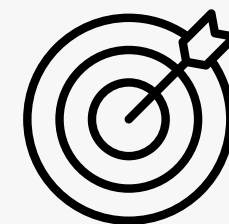


Questions:

In urbans what are the best places to install ads?

Which of these billboards is more visible to pedestrians?

How can we find the best locations for different services in urbans?



Goals:

- Find best locations for different urban functions (Stations, Postboxes,..)
- Placement of billboards and advertisements.
- Allocating different building functions according to user behavior and visual perception.

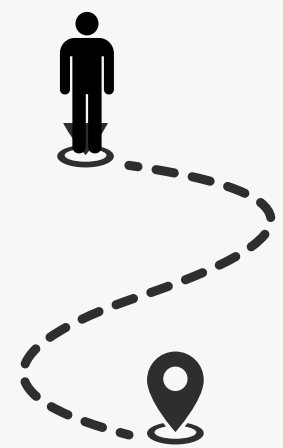


METHODOLOGY

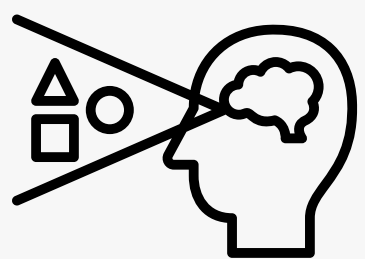
We are answering to these questions by considering the **visual behavior** of humans/agents in public places by:



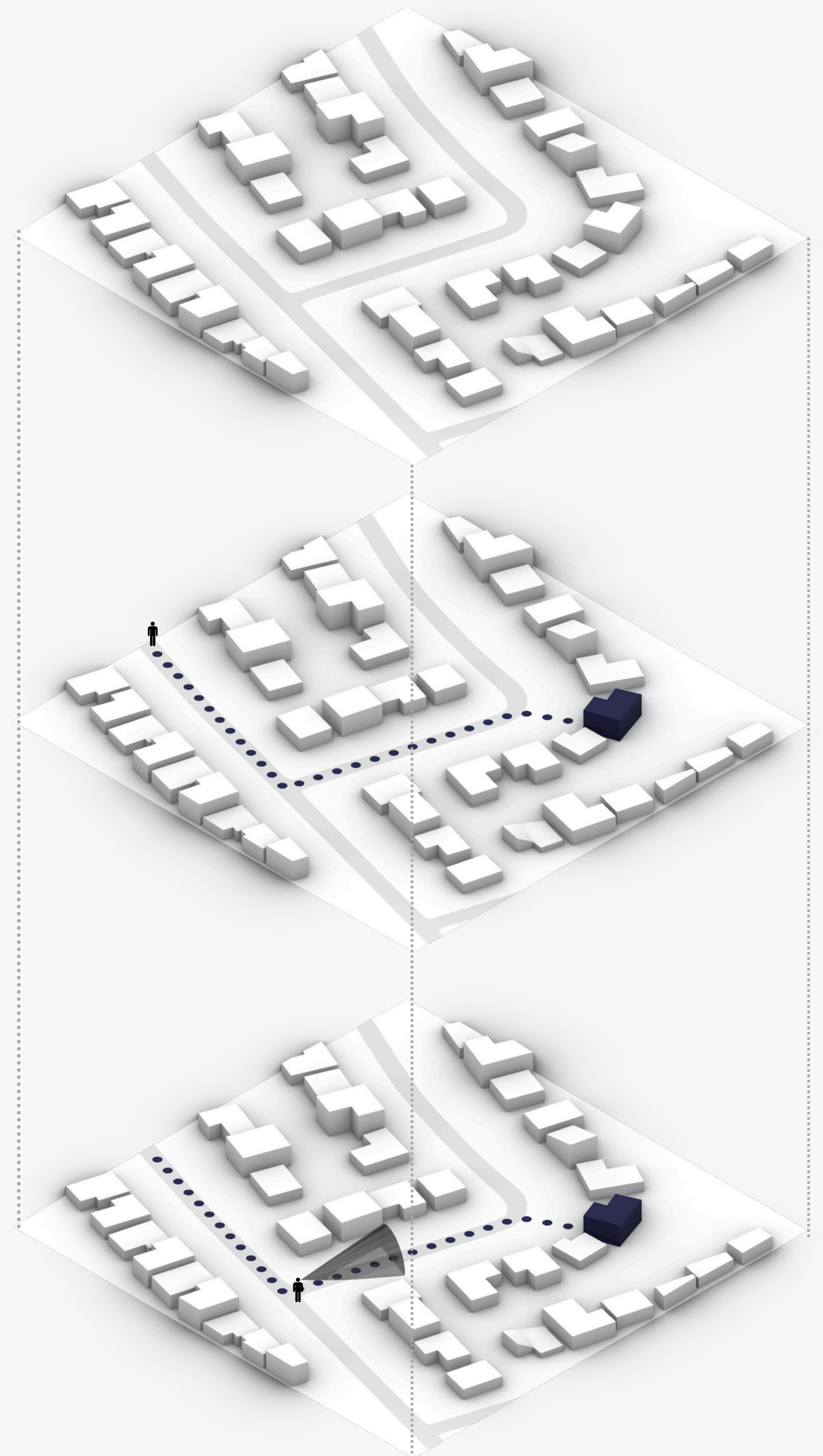
Modeling of simplified urban environment



Path finding based on start and destination



Cone of vision calculation



ALGORITHM DESCRIPTION

1.Urban Context

- Selection of the urban context
- selecting start and destination
- Creating visual objects and barriers
- Finding the shortest path between origin and destination with barriers

2.Agent Navigation

- Dividing the path to specific numbers of steps
- Putting Agent on the origin point
- Selecting visual objects by Agent
- Extraction of normal vector tangent to the Agent's path

3.Cone of Vision

- Specify the end point of line of sight with normal vector and determined distance
- Creating the vertical plane at the end of the line of sight
- Creation of visual curve boundary on the plane

- Making a grid point on the plane and removing the points out of boundary
- Creating the rays from Agent's location point to plane

4.Rays and Plane Intersects

- Extraction of intersect points of rays and plane
- Calculation the first intersection
- Recording the point intersects on objs

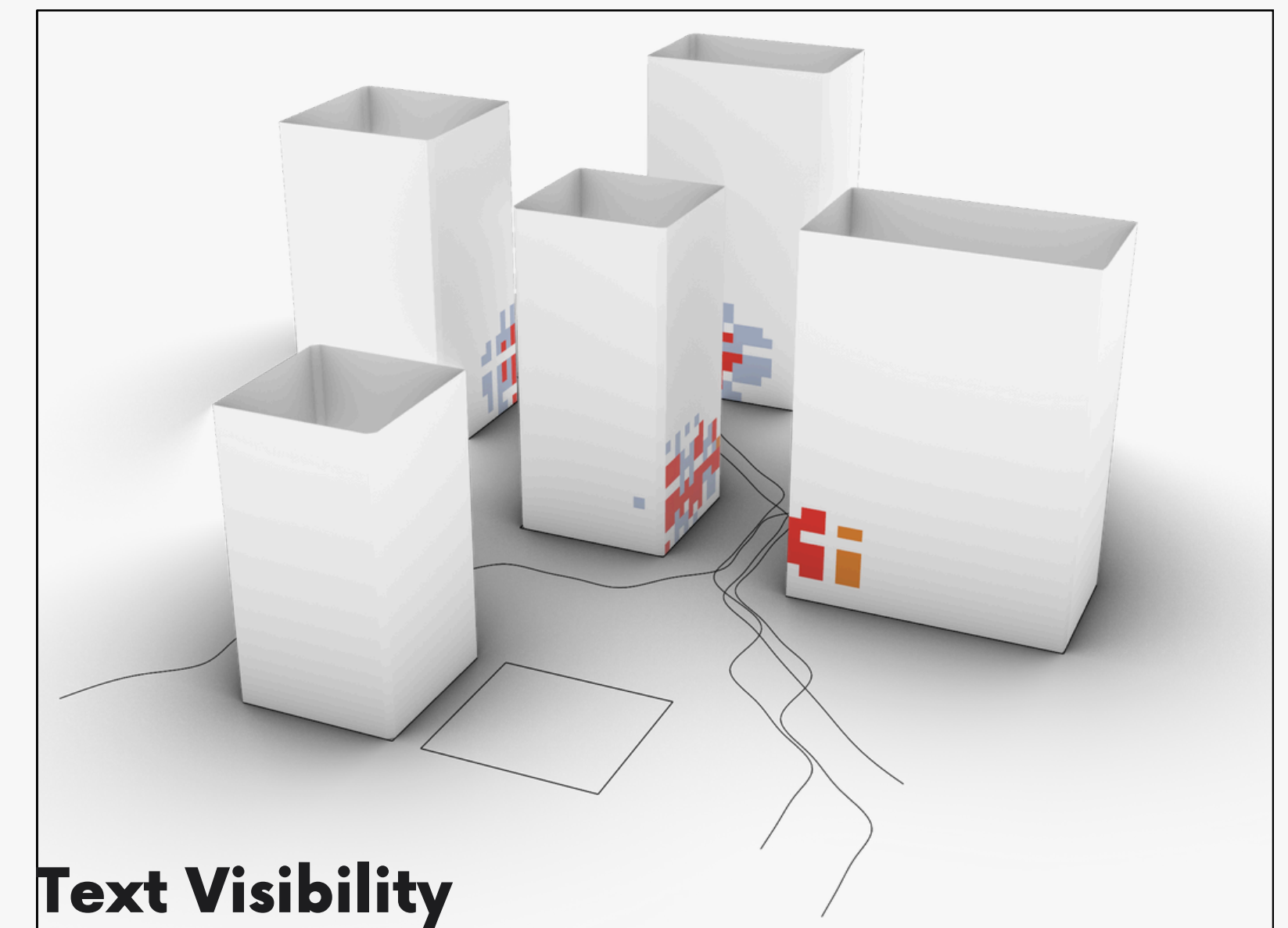
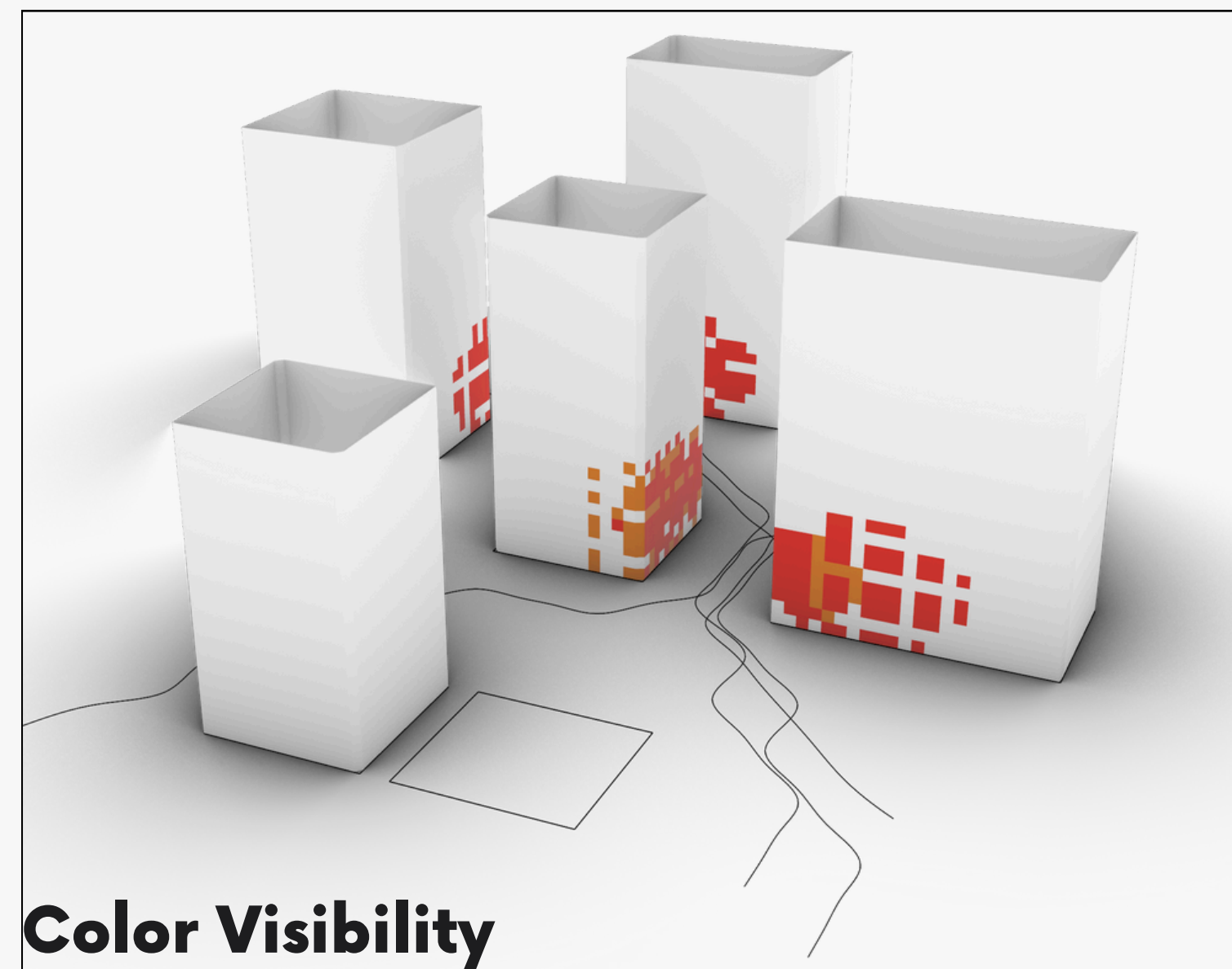
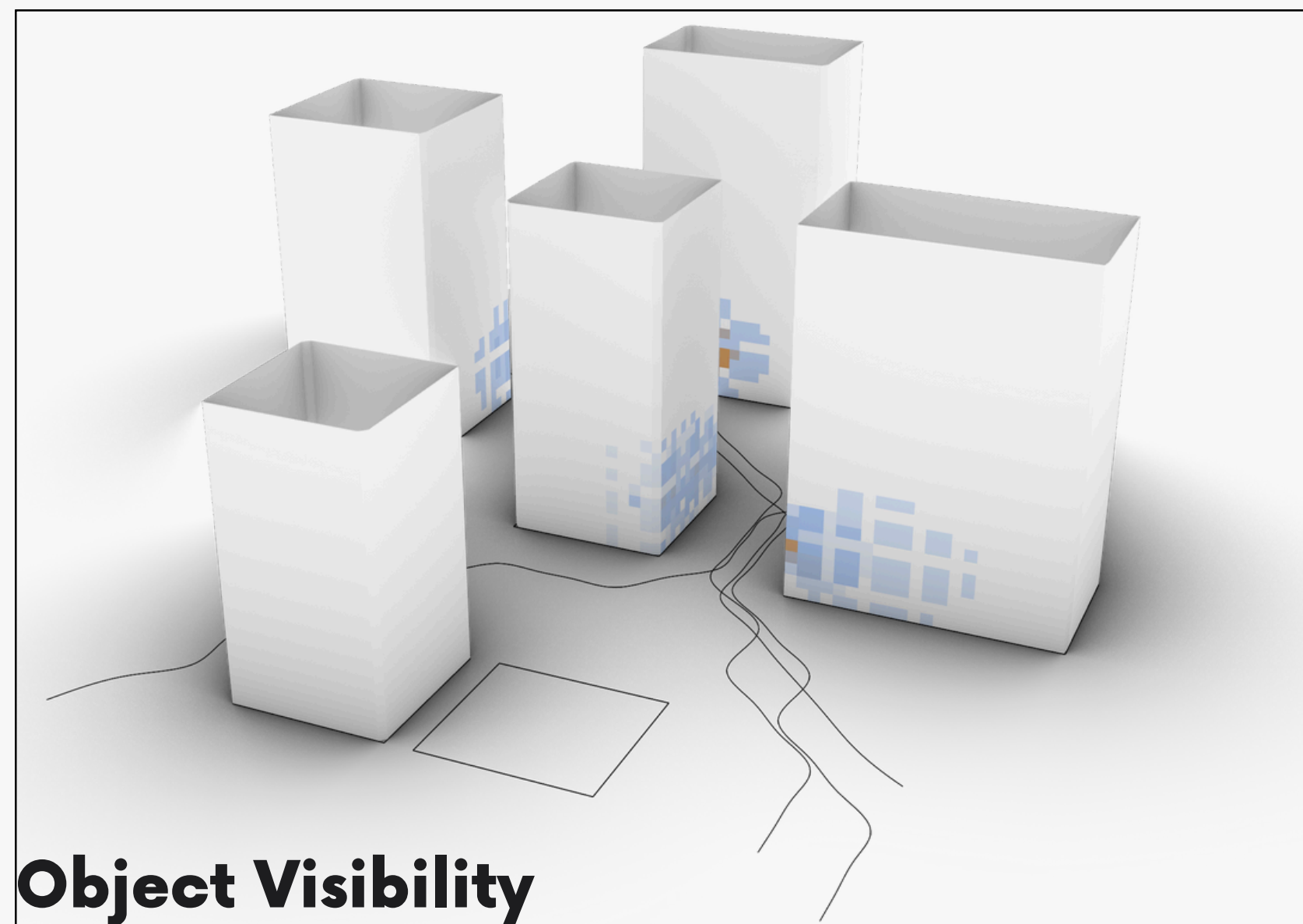
5.Heatmap Visualization

- Movement of Agent in the path and recording Data in each step
- Building grids on the visualized objects planes
- Investigation of number of intersects in each pixel
- Make a list of grid pixels with number of intersections
- Assigning color to pixels

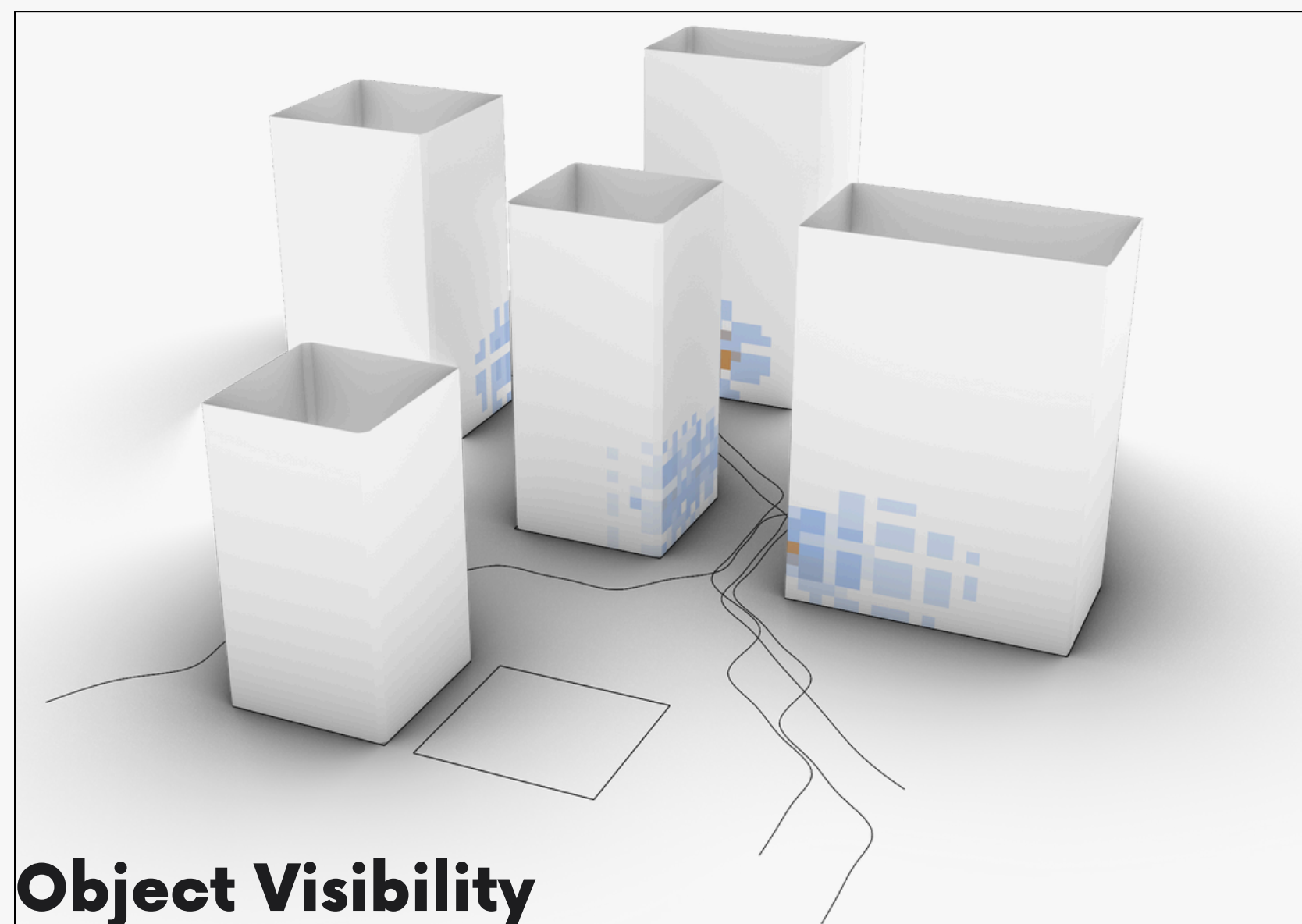
OUTPUTS VISUALIZATION

Classification of sections viewed by Agents to :

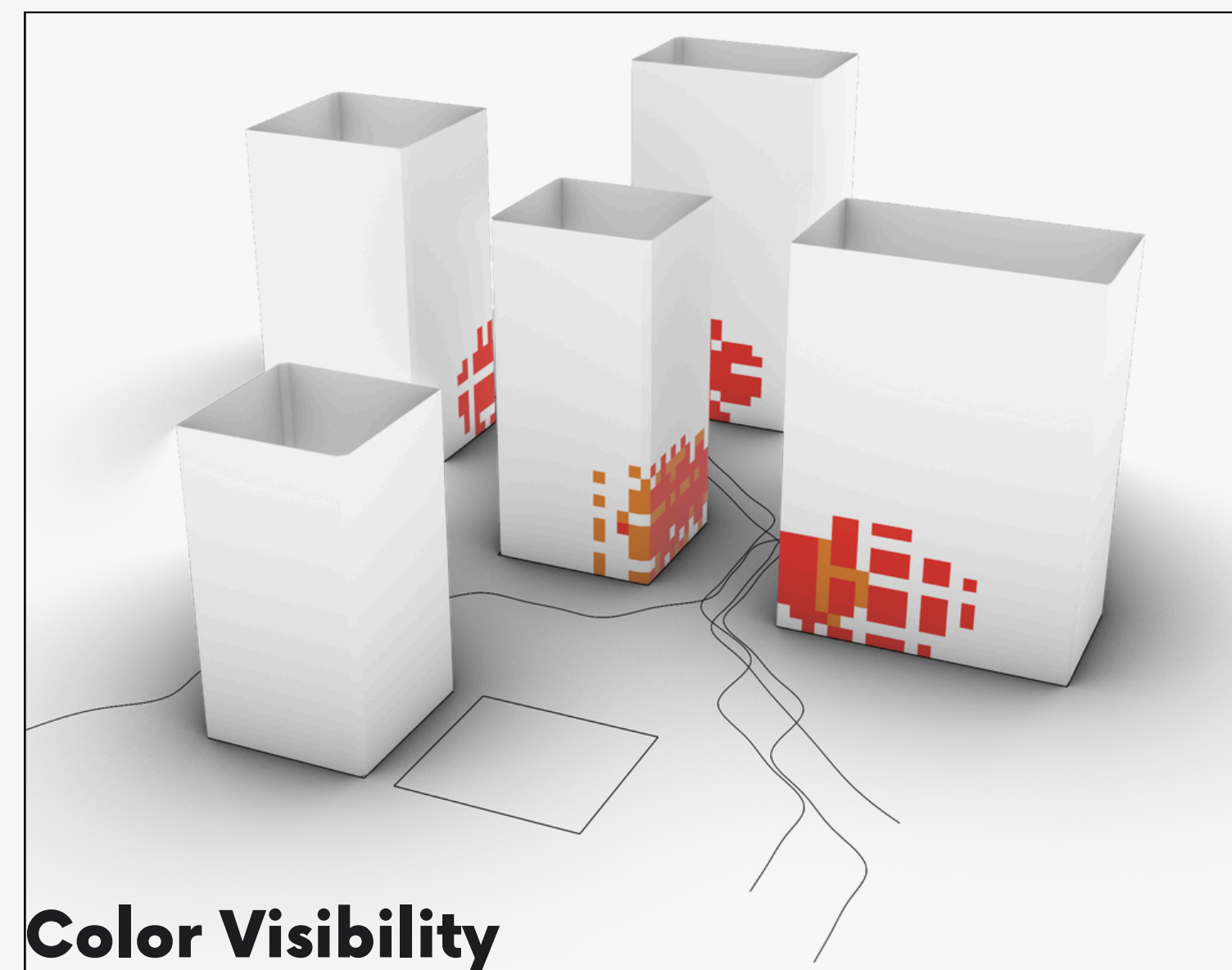
- Object Visibility
- Color Visibility
- Text Visibility



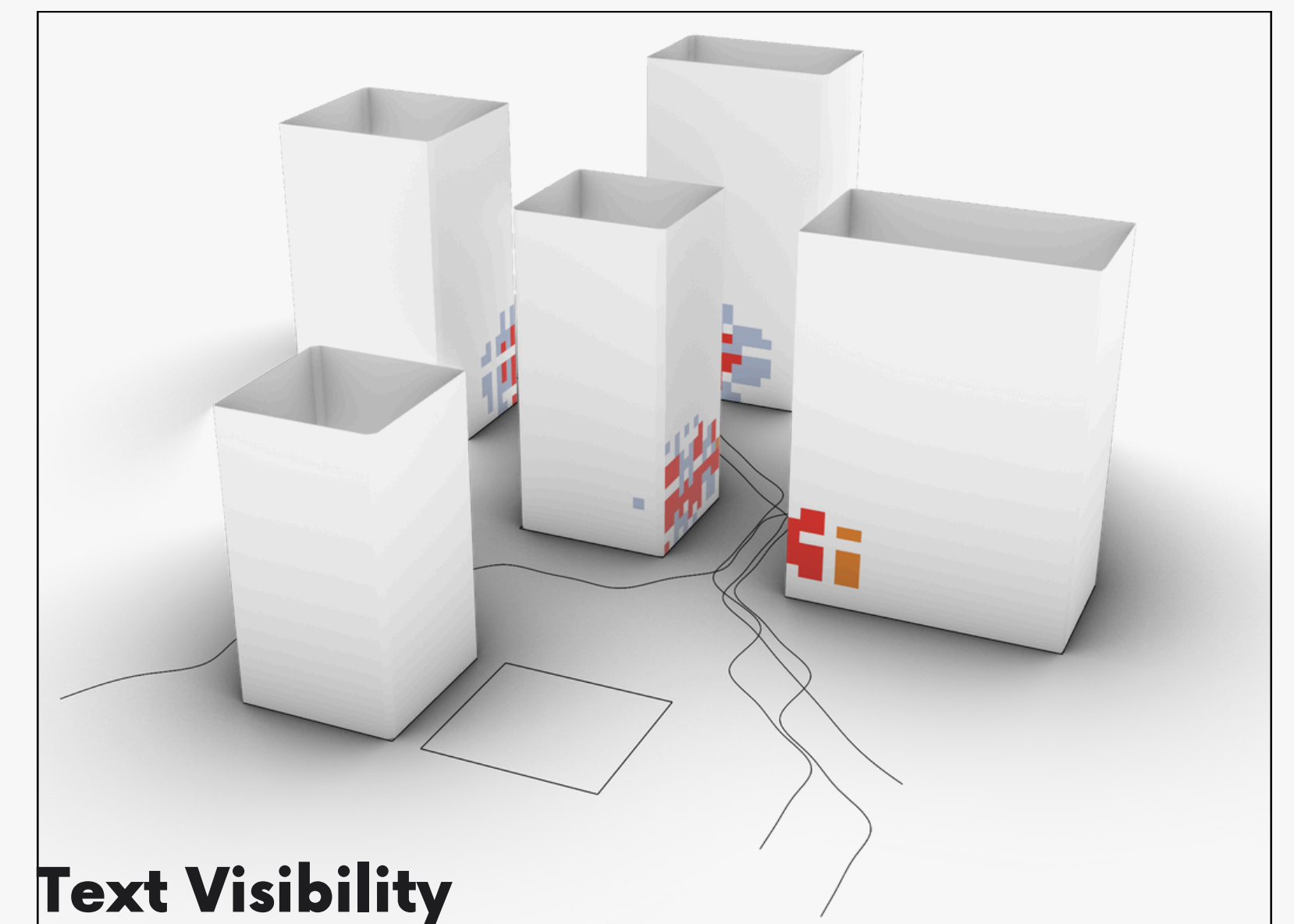
APPLICATION OF RESEARCH



- The exposure level of urban spaces



- Surfaces where color stands out



- Prime spots for ads and text announcements

CHALLENGES & LIMITATIONS

- Lack of awareness of agents behavior
- Different speeds of agents in public areas
- Other intervening variables in urbans like traffic, spatial selection,..

FURTHER STUDIES

- Calculation of Collision Avoidance for more numbers of agents and the possibility of collision
- Calculation Based on Different Users pedestrians and passengers

THANK YOU